Practice

1. A soccer player kicks a 0.43 kg soccer ball down a smooth (frictionless) hill 18 m high with an initial speed of 7.4 m/s (**Figure 5**).
   (a) Calculate the ball's speed as it reaches the bottom of the hill. [ans: 2.0×10^1 m/s]
   (b) The soccer player stands at the same point on the hill and gives the ball a kick up the hill at 4.2 m/s. The ball moves up the hill, comes to rest, and rolls back down the hill. Determine the ball's speed as it reaches the bottom of the hill. [ans: 19 m/s]
2. A tennis player begins a serve by tossing a 57 g tennis ball straight up.
   (a) After leaving the player's hand, the ball rises another 1.8 m. Calculate the speed of the ball as it leaves the player's hand. [ans: 5.9 m/s]
   (b) On the next serve, the tennis player tosses the ball with $\frac{1}{4}$ the speed in (a). Determine the ratio of the maximum rise of the ball after leaving the player's hand after this toss to the maximum rise in (a). [ans: 1:16]

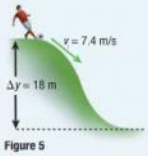

Figure 5

$$E_{gi} + E_{ki} = E_{gf} + E_{kf}$$

$E_g = mg\Delta y$

$E_k = \frac{mv^2}{2}$

1 a) $E_g = 0.43(9.8)\,18$

$E_{g_1} = 75.852$

$E_{k_1} = \frac{0.43\,(7.4)^2}{2} = 11.773$

$E_{T_1} = 75.852 + 11.773 = 87.625$

$E_{gf} = 0$

$E_{kf} = 87.625$

$87.625 = \frac{mv^2}{2}$

$\sqrt{\frac{2(87.625)}{0.43}} = v$

$v = 20.2$ m/s

b) $E_{g_1} = 75.852$

$E_{k_1} = \frac{0.43(4.2)^2}{2}$

$E_{k_1} = 3.792$

$E_{kf} = 3.792 + 75.852$

$E_{gf} = 0$ $E_{nf} = 79.64$ J

$79.64 = \frac{mv^2}{2}$

$v = \sqrt{\frac{79.64(2)}{0.43}}$

$v = 19.2$ m/s

Practice

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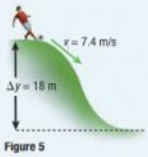

Figure 5

57 g = 0.057 kg

2) $E_{gf} = 1.8(9.8)(0.057)$

$E_{gf} = 1.00548$ J $E_{kf} = 0$

$E_{g_1} = 0$ $E_{k_1} = 1.005$ J

$\frac{mv^2}{2} = 1.005$

$v = \sqrt{\frac{1.005(2)}{0.057}}$

$v = 5.94$ m/s

* Assuming 1.8 m is the highest point *

b) $v_2 = \frac{5.94}{4} = 1.48$ m/s

$E_{gf} = E_{k_1}$

$g(m)(\Delta y) = \frac{mv^2}{2}$

$2g\Delta y = v^2$

$v = \sqrt{2g\Delta y}$

$v_1 = \sqrt{2g\Delta y_1}$

$v_2 = \sqrt{2g\Delta y_2}$

$v_2 = \frac{1}{4} v_1$

$v_1 = 4 v_2$

$\frac{v_1}{v_2} = \frac{\sqrt{2g\Delta y_1}}{\sqrt{2g\Delta y_2}}$

$$= \sqrt{\frac{2g\Delta y_1}{2g\Delta y_2}}$$

$$\frac{v_1}{v_2} = \sqrt{\frac{\Delta y_1}{\Delta y_2}}$$

$$4^2 = \frac{\Delta y_1}{\Delta y_2}$$

$$16 = \frac{\Delta y_1}{\Delta y_2} \qquad y_2 : y_1$$

$$\frac{\Delta y_2}{\Delta y_1} = \frac{1}{16}$$

$$\Delta_2 = 16\Delta y_1$$

$$1 : 16$$

$$P = \frac{W}{t}$$

Practice

1. A firefighter climbs a ladder at a speed of 1.4 m/s. The ladder is 5.0 m long, and the firefighter weighs 65 kg. T/I A
   (a) Determine the firefighter's power output while climbing the ladder. [ans: 890 W]
   (b) How long does it take her to climb the ladder? [ans: 3.6 s]
2. A Grand Prix race car accelerates to twice the speed of the car in Sample Problem 1, in the same amount of time. Calculate the ratio of the power needed by the Grand Prix car to the power needed by the car in Sample Problem 1. K/U T/I A [ans: 4:1]
3. Every year, the Calgary Tower hosts a foot race to the top of the tower. The vertical distance travelled up the 802 steps is about 190 m, and a champion racer can make the climb in less than 5.0 min. If a 62 kg racer completes the climb in 4 min 50 s, determine his average power output during the race. T/I A [ans: 0.40 kW]

1a) $v = 1.4\ m/s$

$\Delta d = 5$

$m = 65\ kg$

$t = \frac{5}{1.4} = \boxed{3.57\ s}$ b

$W = E_k + E_g$

$E_{gf} = 5(9.8)65$

$= 3185$

$\Delta E_g = 3185$

$P = \frac{3185}{3.57} = \boxed{892.2\ W}$ a

2)

Car 1

$W = 4.251 \times 10^5\ J$

$W = \frac{1}{2} m v_f^2$

$\Delta t = 7.7\ s$

$V_i = 0$

$V_f = 27.8\ m/s$

Car 2

$W = \frac{1}{2} m \cdot v_f^2$

$\Delta t = 7.7\ s$

$P = \frac{W}{t}$

$V_2 = 2V_1$

$W_1 = \frac{1}{2} m V_1^2$

$V_1 = \sqrt{\frac{2W_1}{m}}$

$W_2 = \frac{1}{2} m V_2^2$

$V_2 = \sqrt{\frac{2W_2}{m}}$

$$\frac{V_2}{V_1} = \frac{2}{1}$$

$$\frac{P_2}{P_1} = \frac{V_2^2}{V_1^2}$$

$$P_1 = \frac{mv_1^2}{2t}$$

$$P_2 = \frac{mv_2^2}{2t}$$

$$\frac{P_1}{P_2} = \frac{\frac{mv_1^2}{2t}}{\frac{mv_2^2}{2t}} = \frac{V_1^2}{V_2^2}$$

$$\frac{P_1}{P_2} = \frac{V_1^2}{V_2^2}$$

$$2V_1^2 = V_2^2$$

Sample Problem 1: Power as a Rate of Change in Kinetic Energy

A car accelerates from rest to a speed of 27.8 m/s in 7.7 s. The mass of the car is 1.1×10^3 kg. Ignoring friction, determine how much power the car requires.

Given: $m = 1.1 \times 10^3$ kg; $v_i = 0$ m/s; $v_f = 27.8$ m/s; $t = 7.7$ s

Required: P

Analysis: Use the power equation, $P = \frac{W}{t}$. Use the work–energy theorem to relate W to the change in kinetic energy ΔE_k.

Solution: The work done on the car equals its change in kinetic energy. The car starts from rest, so the change in kinetic energy equals the final kinetic energy:

$$\begin{aligned} W &= \Delta E_k \\ &= E_{k_f} \\ &= \frac{1}{2}mv_f^2 \\ &= \frac{1}{2}(1.1 \times 10^3\ \text{kg})(27.8\ \text{m/s})^2 \end{aligned}$$

$W = 4.251 \times 10^5$ J (two extra digits carried)

Therefore,

$$\begin{aligned} P &= \frac{W}{t} \\ &= \frac{4.251 \times 10^5\ \text{J}}{7.7\ \text{s}} \\ P &= 5.5 \times 10^4\ \text{W} \end{aligned}$$

Statement: The car requires 55 kW of power.

$$\frac{P_1}{P_2} = \frac{1V_1^2}{2V_1^2}$$

$$\frac{P_1}{P_2} = \frac{1^2}{2^2}$$

$4P_2 : 1P_1$

$4 : 1$

3. Every year, the Calgary Tower hosts a foot race to the top of the tower. The vertical distance travelled up the 802 steps is about 190 m, and a champion racer can make the climb in less than 5.0 min. If a 62 kg racer completes the climb in 4 min 50 s, determine his average power output during the race. [ans: 0.40 kW]

4 min = 240 s

$\Delta t = 290\,s$

$m = 62\,kg$

$\Delta d = 190\,m$

$W = E_g$

$E_g = mgh$

$P = \frac{62(9.8)(190)}{290}$

$P = 398\,W$

$P = 0.4\,kW$

1. A child tosses a tennis ball straight up into the air with an initial speed of 11 m/s. Ignore the height of the child, and assume that air resistance is negligible.
   (a) Determine the maximum height that the ball will reach.
   (b) Sketch a graph of potential energy versus time for the ball during its time in the air. Explain why the graph has the shape that it does. Label the minimum and maximum potential energy values.
   (c) On the same set of axes, sketch a graph of total energy versus time and a graph of kinetic energy versus time. Explain why they are shaped the way that they are. Identify any maximum or minimum values.
2. An apple falls from a branch to the ground below.
   (a) At what moment is the kinetic energy of the apple greatest?
   (b) At what moment is the gravitational potential energy greatest?
3. A hockey puck slides along a level surface, eventually coming to rest.
   (a) Is the energy of the hockey puck conserved? Explain your answer.
   (b) Discuss what happens to the initial kinetic energy of the puck.
4. (a) A skier of mass 110 kg travels down a frictionless ski trail with a top elevation of 210 m. Calculate the work done on the skier by gravity as the skier travels from the top of the trail to the bottom.
   (b) Calculate the speed of the skier when he reaches the bottom of the ski trail. Assume he starts from rest.
5. A 62 kg snowboarder is moving across a horizontal ledge at 8.1 m/s when she encounters a drop-off and becomes airborne. Ignore air resistance. The snowboarder lands 3.7 m below the drop-off. Calculate her speed at the moment she hits the ground.
6. A dolphin is trying to jump through a hoop that is fixed at a height of 3.5 m above the surface of her pool. The dolphin leaves the water at an angle of inclination of 40°. Determine the minimum speed the dolphin will need when leaving the water in order to reach the height of the hoop.
7. A roller coaster car with mass 640 kg moves along the track shown in **Figure 8**. Assume all friction is negligible.

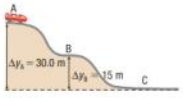

Figure 8

   (a) Is the mechanical energy of the roller coaster conserved? Explain your answer.
   (b) If the roller coaster starts from rest at point A, what is its total mechanical energy at point A?
   (c) What is the total mechanical energy at point B?
   (d) Calculate the speed of the roller coaster when it reaches points B and C.
   (e) If the car starts with a speed of 12 m/s at point A, calculate the speed of the roller coaster when it reaches points B and C.
8. A 52 kg woman jogs up a hill in 24 s. Calculate the power the woman exerts if the hill is 18 m high.

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1 a) $V_1 = 11\,m/s$

$V_f = 0$

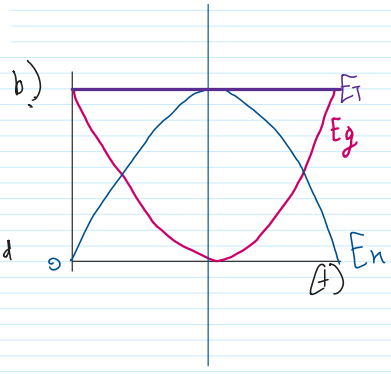
$V_f^2 = V_i^2 + 2\vec{a}\Delta d$

$0 = 11^2 + 2(-9.8)\Delta d$

$\frac{-121}{\Delta d} = -19.6$

$\Delta d = \frac{121}{19.6}$

b)



$\frac{121}{\Delta d} = 19.6$

$\Delta d = \frac{121}{19.6}$

$\Delta d = 6.17 m$

2. An apple falls from a branch to the ground below.
   (a) At what moment is the kinetic energy of the apple greatest?
   (b) At what moment is the gravitational potential energy greatest?
3. A hockey puck slides along a level surface, eventually coming to rest.
   (a) Is the energy of the hockey puck conserved? Explain your answer.
   (b) Discuss what happens to the initial kinetic energy of the puck.

2 a) Right before it hits the ground

b) Right as it falls

3 a) If the puck and the surface form an isolated system, yes energy is conserved. The energy is converted to thermal energy by friction. So it's not lost, just converted.

4. (a) A skier of mass 110 kg travels down a frictionless ski trail with a top elevation of 210 m. Calculate the work done on the skier by gravity as the skier travels from the top of the trail to the bottom.
   (b) Calculate the speed of the skier when he reaches the bottom of the ski trail. Assume he starts from rest.

$\Delta y = 210 m$

$m = 110 kg$

$W = ?$

$W = E_g$

$E_{g_1} = mg\Delta y$

$= 110(9.8)210$

a) $= 226,380 J$

[The work done by gravity is = to the gravitational Potential energy]

b) $E_{g_1} = E_{kf}$

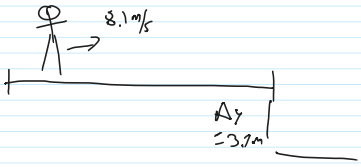
$E_{kf} = \frac{mv^2}{2}$

$226,380 = \frac{110(v)^2}{2}$

$\sqrt{\frac{226,380(2)}{110}} = v$

$v = 64.156 m/s$

5. A 62 kg snowboarder is moving across a horizontal ledge at 8.1 m/s when she encounters a drop-off and becomes airborne. Ignore air resistance. The snowboarder lands 3.7 m below the drop-off. Calculate her speed at the moment she hits the ground.



$E_{T_1} = E_{g_1} + E_{k_1}$

$E_{g_1} = 62(9.8)\ 3.7 = 2248.12$

$E_{k_1} = \frac{62(8.1)^2}{2} = 2033.91$

$E_{T_1} = 4282.03$

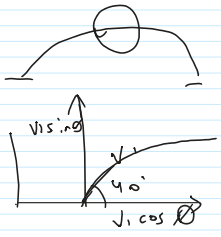
$E_{g_2} = 0$

$E_{k_2} = 4282.03$

$v = \sqrt{\frac{4282.03 \times 2}{62}}$

$v = 11.75 m/s$

6. A dolphin is trying to jump through a hoop that is fixed at a height of 3.5 m above the surface of her pool. The dolphin leaves the water at an angle of inclination of 40°. Determine the minimum speed the dolphin will need when leaving the water in order to reach the height of the hoop.



$E_{gf} = m(9.8)(3.5)$

$E_k = m(v_1 \sin\theta)^2$

$v_1 \cos\theta$

$E_{gf} = m(9.8)(3.5)$

$E_{k1} = \frac{m(v_1 \sin\theta)^2}{2}$

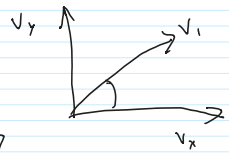
$v_y = v_1 \sin\theta$

$\cancel{m}(9.8)(3.5) = \cancel{m}(v_y)^2 (\frac{1}{2})$

$v_y = \sqrt{(9.8)(3.5)(2)}$

$v_y = 8.28 m/s$

v_{1y} = vertical



[The energy equation gives us speed in the direction of the jump]

$v_y = v_1 \sin 40$

$\frac{v_y}{\sin 40} = v_1$

$v_1 = \frac{8.28}{\sin 40}$

$v_1 = 12.881 m/s$

7. A roller coaster car with mass 640 kg moves along the track shown in **Figure 8**. Assume all friction is negligible.

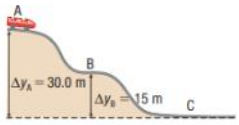


Figure 8

(a) Is the mechanical energy of the roller coaster conserved? Explain your answer.
(b) If the roller coaster starts from rest at point A, what is its total mechanical energy at point A?
(c) What is the total mechanical energy at point B?
(d) Calculate the speed of the roller coaster when it reaches points B and C.
(e) If the car starts with a speed of 12 m/s at point A, calculate the speed of the roller coaster when it reaches points B and C.

7) a) Yes. There is no friction so the energy is conserved.

b) $E_T = E_g$

$= 640(9.8)30$

$E_T = 188160 J$

c) $E_T = 188160 J$

d) B:

$E_g = 640(9.8)(15) = 94080$

$E_k = 188160 - 94080$

$= 94080$

$v = \sqrt{\frac{(2)94080}{640}}$

$v = 17.146 m/s$

C:

$E_k = 188160$

$v = \sqrt{\frac{2(188160)}{640}}$

$v = 24.248 m/s$

e) A: $E_k = \frac{640(12)^2}{2} = 46,080$

$E_t = 188160 + 46080$

$= 234240$

B: $E_k = 234240 - 94080$

$= 140160 J$

$v = \sqrt{\frac{2(140160)}{640}}$

$v = 20.928$

C: $E_k = 234,240$

$v = \sqrt{\frac{2(234,240)}{640}}$

$v =$

$v = 27.055 m/s$

8. A 52 kg woman jogs up a hill in 24 s. Calculate the power the woman exerts if the hill is 18 m high.

8) $m = 52$

$\Delta t = 24$

$\Delta y = 18 m$

$P = \frac{W}{\Delta t}$

$W = \Delta E$

$\Delta E = E_{gf} = 52(9.8)(18)$

$= 9172.8$

$$P = \frac{9172.8}{24}$$

$$P = 382.2\ W$$